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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

ME1101-1- ENGINEERING METROLOGY
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the approximate size of slip gauges?**
A) 30mm long and 10mm wide B) 45mm long and 15mm wide
C) 20mm long and 5mm wide D) 25mm long and 10mm wide
2. **Which of the following is not correct about wavelength standard?**
A) Wavelength of monochromatic light is an invariable unit of length B) Wavelength standard is a physical standard
C) It is a reproducible standard D) It is a natural unit of length
3. **Which of the following is not the most important feature of slip gauge?**
A) Length between measuring surface B) Flatness
C) Surface conditions of measuring surface D) Adhereness efficiency
4. **Which of the following is incorrect about line standard?**
A) These are legal standards B) Slope at the ends is zero
C) A round recess, half inch away from the ends is cut at both ends D) Points of supports are called 'Airy' points
5. **Why a revolution counter is used in dial gauge?**
A) To withstand sudden shock B) To indicate correctly the number of revolution
C) To count and display the number of revolutions D) To transfer motion at right angle
6. **Which of the following is not used in making of sine bars?**
A) High carbon B) High chromium
C) Corrosion resistant steel D) Aluminium
7. **Which of the following is true for uses of comparators?**
A) Can't be used in mass production B) Not suitable for inspection purposes
C) Slow rate of working D) Can be used as working gauge
8. **What is the standard pressure angle for an involute system?**
A) 20 or 30 degree B) 30 or 14.5 degree
C) 40 or 30 degree D) 20 or 14.5 degree
9. **The output of a transducer must _____**
A) be different at different environment conditions B) be same at all environment conditions
C) be same at some environment conditions D) be zero always

10. **Capacitive transduction involves** _____
 A) change in resistance B) change in capacitance
 C) change in inductance D) change in potential
11. **How do passive transducers develop electrical signals?**
 A) using a transformer B) using external source
 C) using internal source D) using a diode
12. **Mechanical transducers cause** _____
 A) power loss B) eddy current loss
 C) hysteresis loss D) frictional loss
13. **Strain gauge measurement involves** _____
 A) Kelvin bridge B) De Sauty's bridge
 C) Wheatstone bridge D) Anderson bridge
14. **Load cell is used for the measurement of** _____
 A) Area B) Force
 C) Length D) Mass
15. **Temperature transducers make use of** _____
 A) change in area B) change in capacitance
 C) change in length D) change in resistivity
16. **Load cells are calibrated such that** _____
 A) force varies directly with resistance B) force varies as the square of resistance
 C) force varies inversely with resistance D) force remains constant with resistance
17. **Upper deviation is the algebraic difference between the _____ and corresponding basic size.**
 A) minimum, maximum B) minimum, minimum
 C) maximum, maximum D) maximum, minimum
18. _____ **is the difference between the size of the manufactured product and the corresponding basic size.**
 A) Deviation B) Upper deviation
 C) Actual deviation D) Allowance
19. **Which of the following option is not correct for 'full interchangeability'?**
 A) This type of interchangeability is not feasible sometimes B) Requires machine which can maintain low process capability
 C) Machines with very high accuracy are necessary D) For interchangeable production, this type of interchangeability is not must
20. **Which of the following option is incorrect about interchangeability?**
 A) Increase output B) Assembly time increases
 C) Increase cost of production D) Useful in mass production

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is metrology? State the objectives of metrology.	08	L*1	1	1
	b) Describe with neat sketch working principle of sine bar and also list the limitations of sine bar.	08	L2	2	1
2.	a) With a neat sketch explain wringing phenomenon. Using M112 slip gauge set build the dimension of 49.3115 mm	08	L3	1	2
	b) Write a note on angle gauges and build an angle of 37° 9' 18" using them.	08	L2	2	1
3.	a) Explain with neat sketch International Prototype Meter. What are the disadvantages of material length standards?	08	L2	1	1

- b) Explain the working principle of optical comparator. With a neat sketch explain Zeiss ultra-optimizer.

08 L3 2 1

Unit – II

4. a) Discuss with block diagram generalized measurement system with examples for each stage.

08 L3 3 1

- b) With the help of a neat sketch, explain the working and construction of a Prony Brake Dynamometer.

08 L3 4 1

5. a) What is active and passive transducers? Explain any one mechanical transducers.

08 L2 3 1

- b) What is a thermocouple? State the laws governing the functioning of thermocouples.

08 L2 4 1

6. a) Explain the working of CRO with the help of a neat sketch.

08 L3 3 1

- b) Explain the construction and working of Berry strain gauges with a neat sketch.

08 L3 4 1

Unit – III

7. a) Differentiate between Hole basis System and Shaft basis system of fits. State the significance of the Hole basis system.

08 L3 5 1

- b) Explain the concepts of interchangeability and selective assembly. Discuss their significance in manufacturing and engineering applications with relevant examples.

08 L3 5 1

8. a) State and explain Taylor's principle of gauge design.

08 L3 5 1

- b) Determine the dimensions of the shaft and hole for a fit $30 H_8/d_{10}$ and sketch the fit, given data:

i) diameter 30 falls in the diameter range 18-30, upper deviation for 'd' shaft is $-16 D^{0.44}$

ii) $i = 0.45(D)^{1/3} + 0.001D$. Tolerance for IT8 = 25 i and Tolerance for IT10 = 64 i.

08 L4 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
